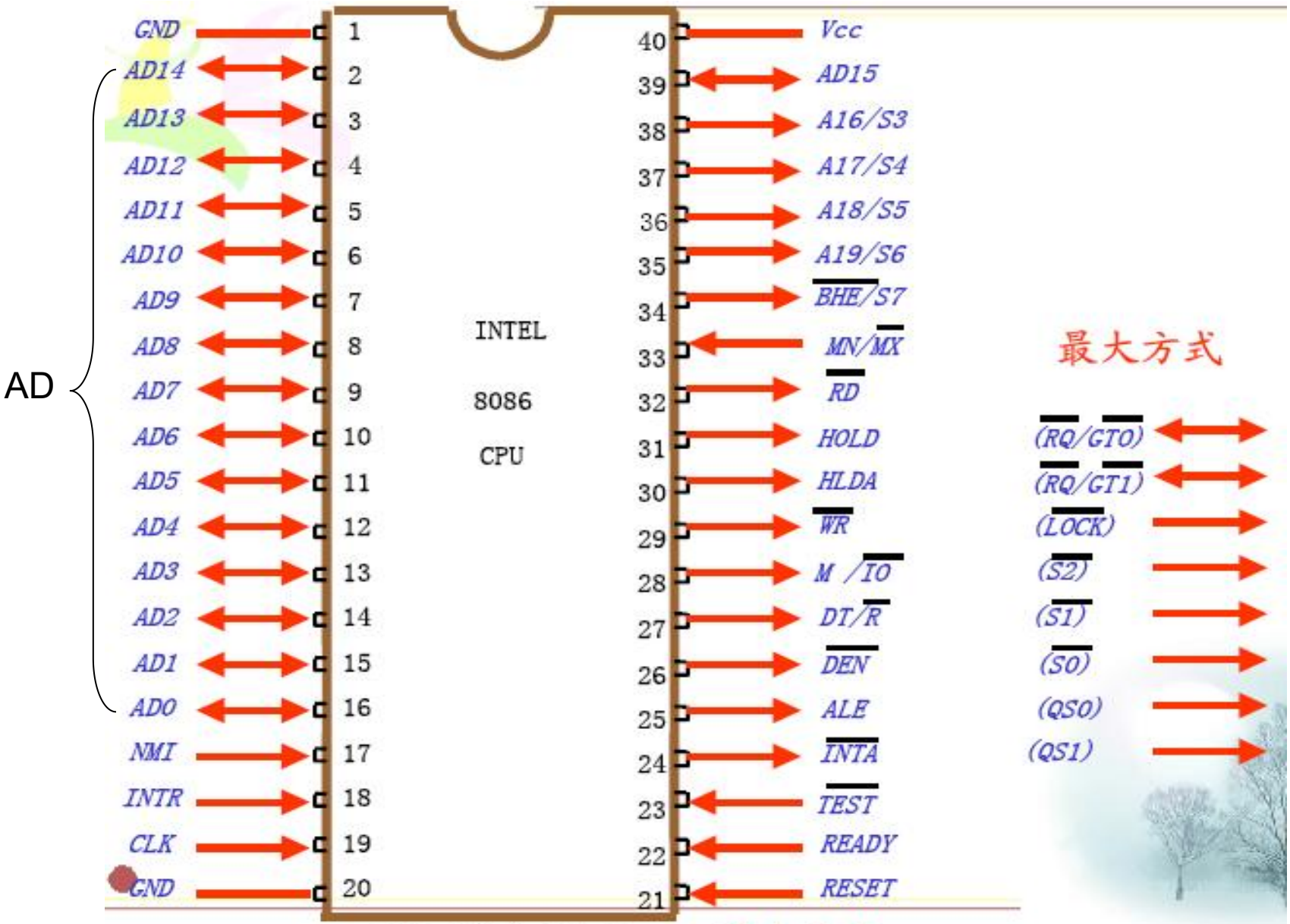
8086/8088 CPU 工作模式


AD
GND 1
AD14 2
AD13 3
AD12 4
AD11 5
AD10 6
AD9 7
AD8 8
AD7 9
AD6 10
AD5 11
AD4 12
AD3 13
AD2 14
AD1 15
AD0 16
NMI 17
INTR 18
CLK 19
GND 20
INTEL
8086
CPU
40 Vcc
39 AD15
38 A16/S3
37 A17/S4
36 A18/S5
35 A19/S6
34 BHE/S7
33 MN/MX
32 RD
31 HOLD
30 HLDA
29 WR
28 M /IO
27 DT/R
26 DEN
25 ALE
24 INTA
23 TEST
22 READY
21 RESET
最大方式
(RQ/GT0)
(RQ/GT1)
(LOCK)
(S2)
(S1)
(S0)
(QS0)
(QS1)
最小方式

8086引脚说明

8086/8088CPU工作方式

当MN/$\overline{MX}$引脚接地时，8086/8088CPU工作于最大模式。

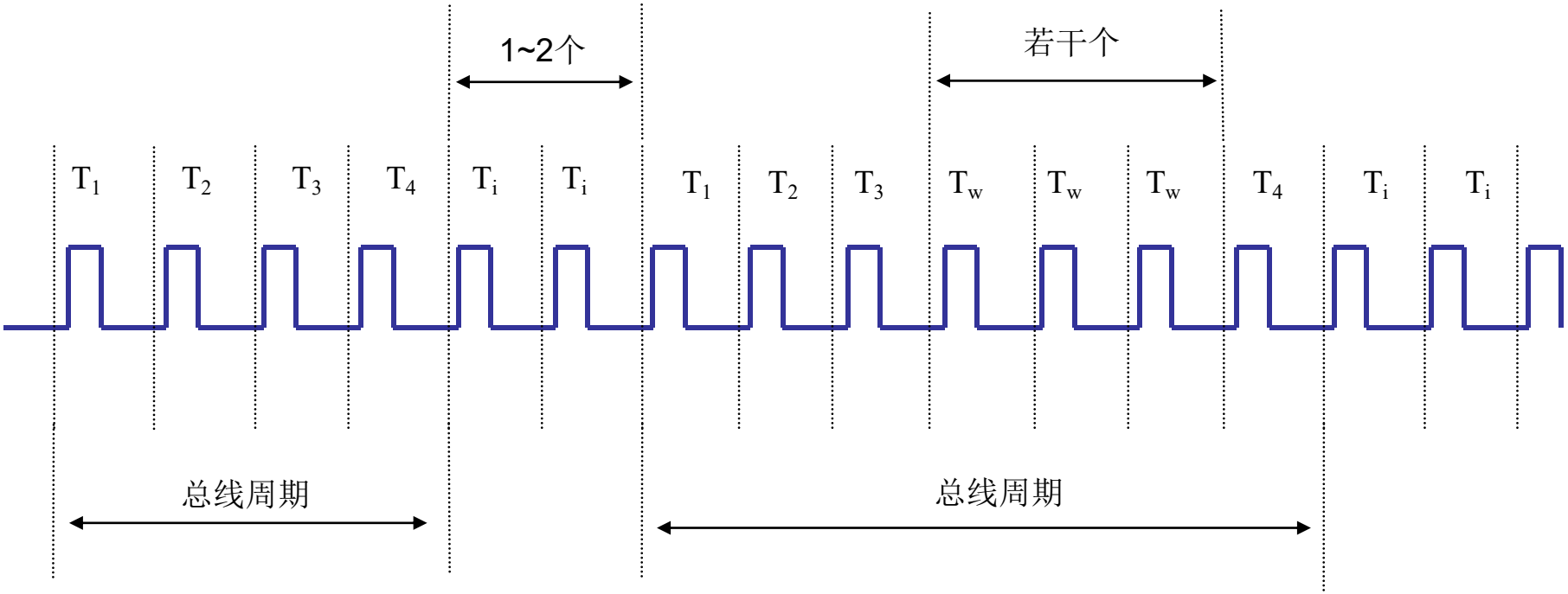
当MN/$\overline{MX}$引脚接+5V时，8086/8088CPU工作于最小模式。

- **最小模式时, 存储器与IO区分:**
  对8086 CPU而言，用控制信号 $M/\overline{IO}$ 来区分，
  对8088 CPU而言，用控制信号 $\overline{M}/IO$ 来区分。

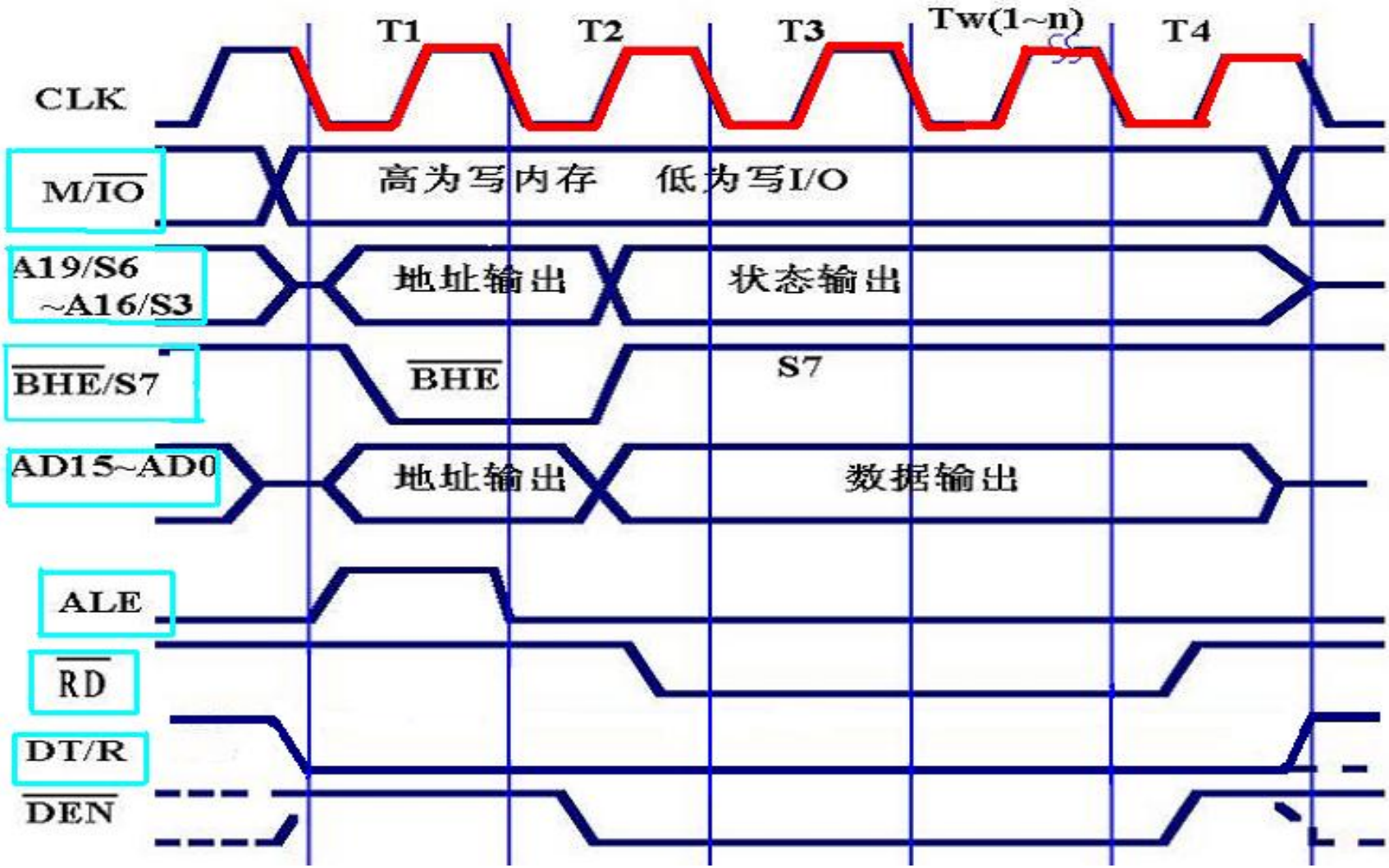
- 8086 CPU AD_0～AD_{15},外部数据线16位，

  8088 CPU AD_0～AD_7,外部数据线8位，

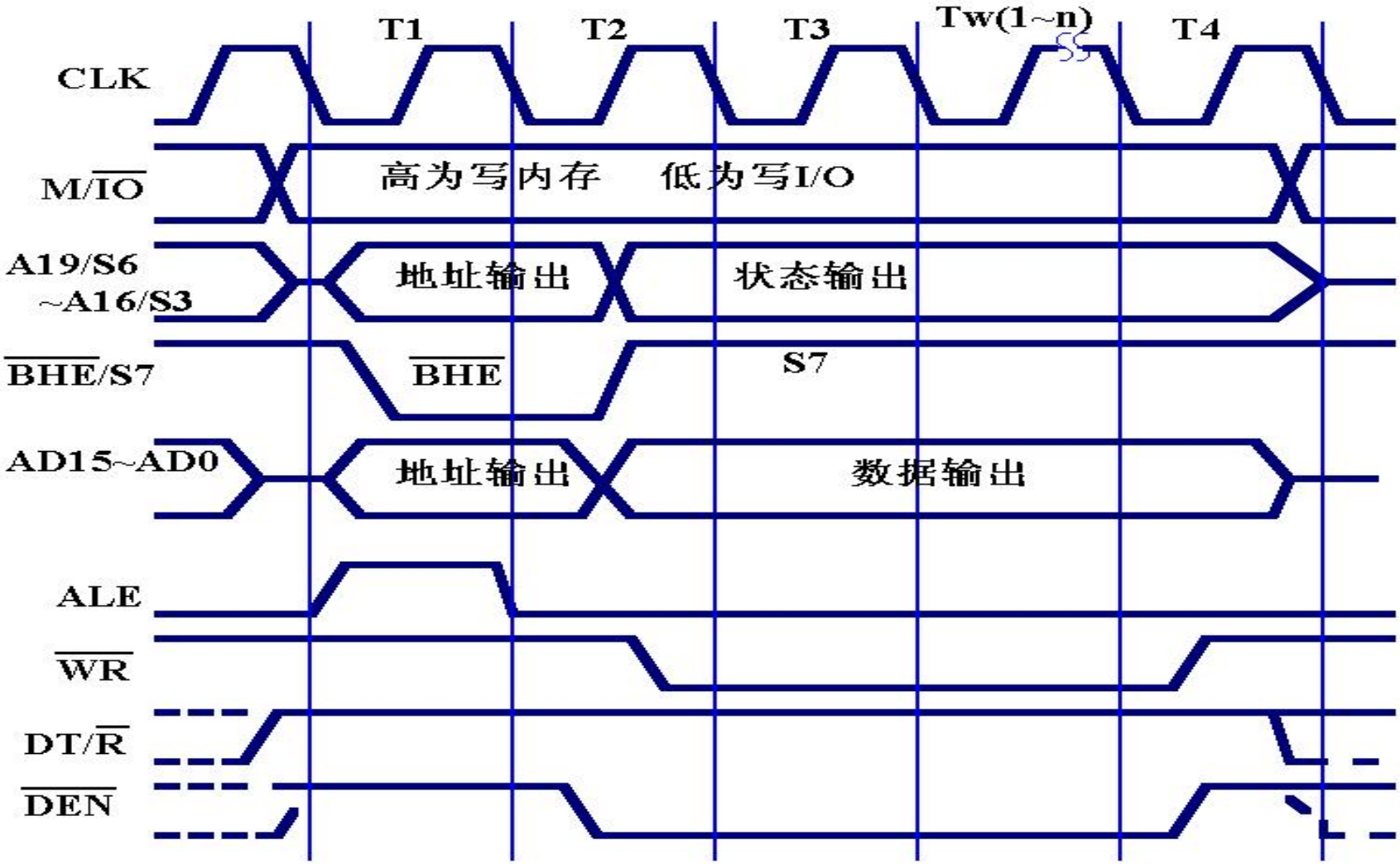
总线周期



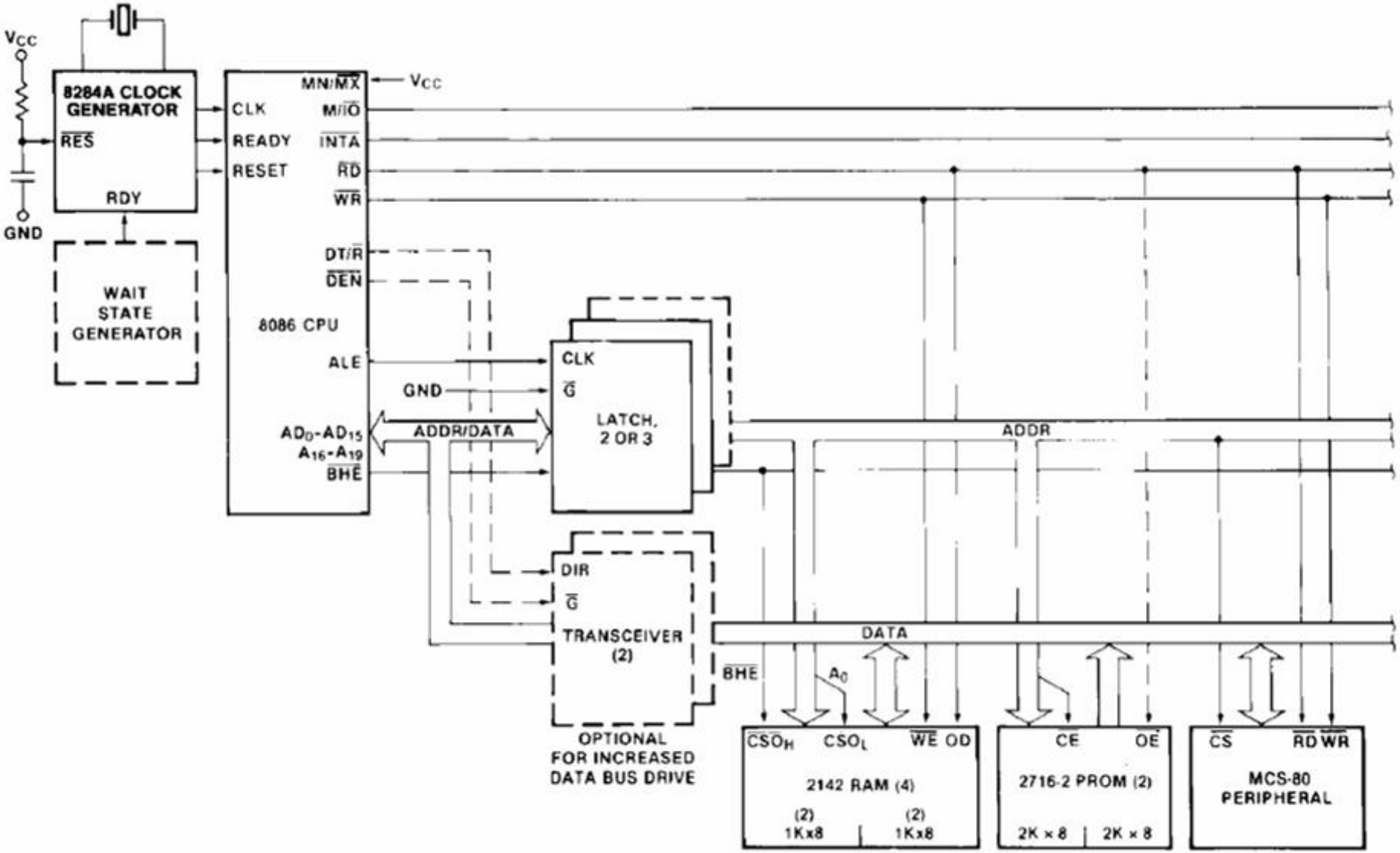
读时序


T1
T2
T3
Tw(1~n)
T4
CLK
M/$\overline{IO}$
高为写内存
低为写I/O
A19/S6
~A16/S3
地址输出
状态输出
$\overline{BHE}$/S7
$\overline{BHE}$
S7
AD15~AD0
地址输出
数据输出
ALE
$\overline{RD}$
DT/R
$\overline{DEN}$

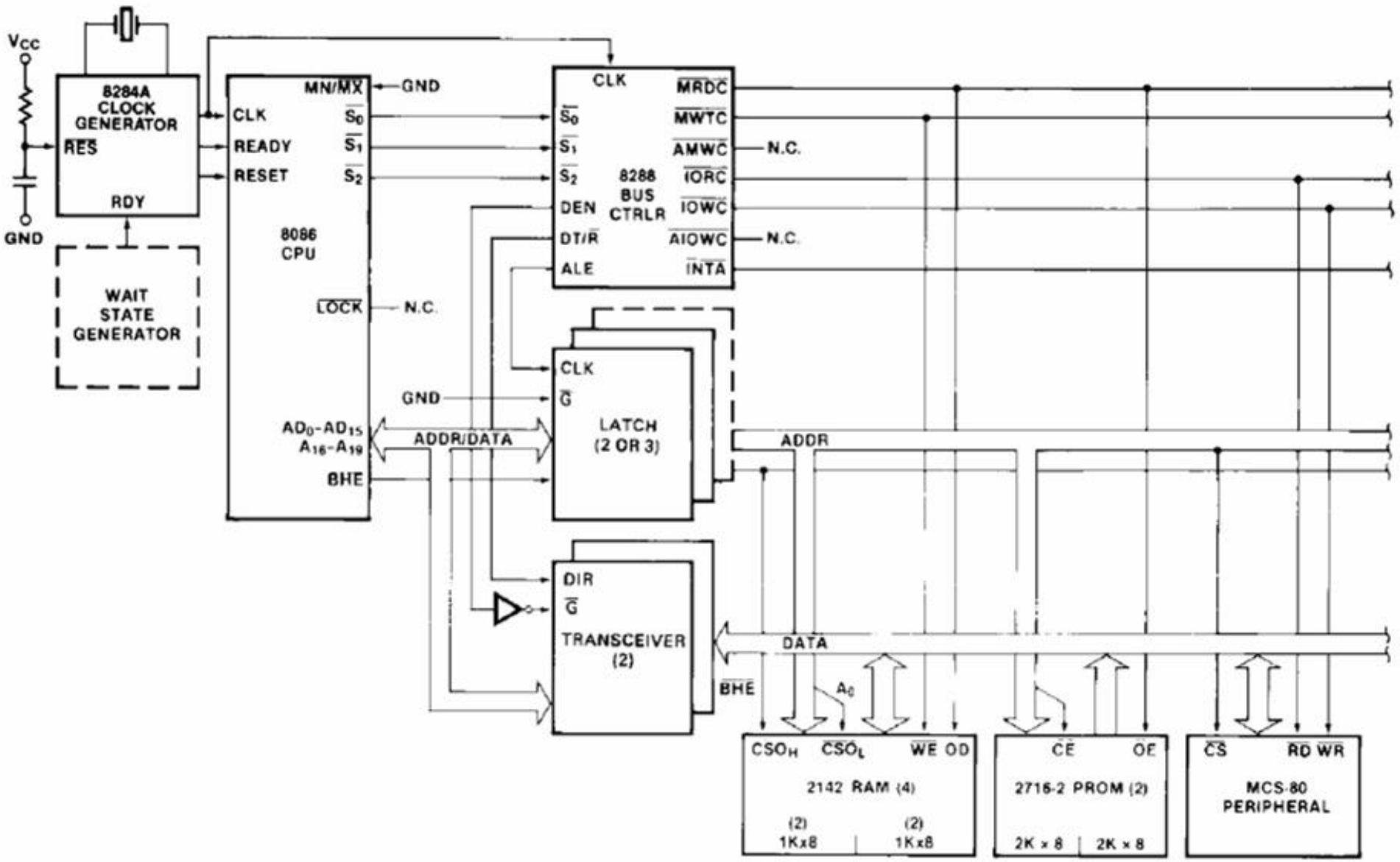
写时序


T1
T2
T3
Tw(1~n)
T4
CLK
M/$\overline{IO}$
高为写内存
低为写I/O
A19/S6
~A16/S3
地址输出
状态输出
$\overline{BHE}$/S7
$\overline{BHE}$
S7
AD15~AD0
地址输出
数据输出
ALE
$\overline{WR}$
DT/$\overline{R}$
$\overline{DEN}$

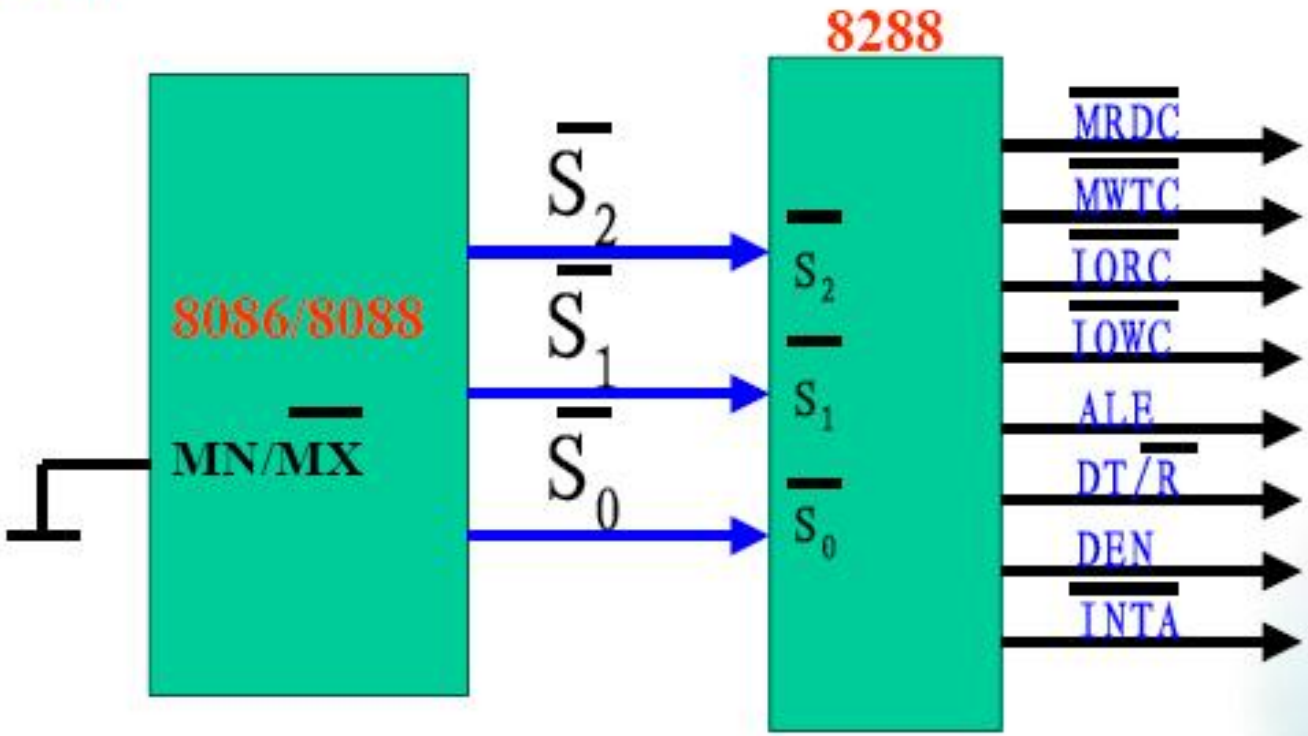
8086 最小工作模式

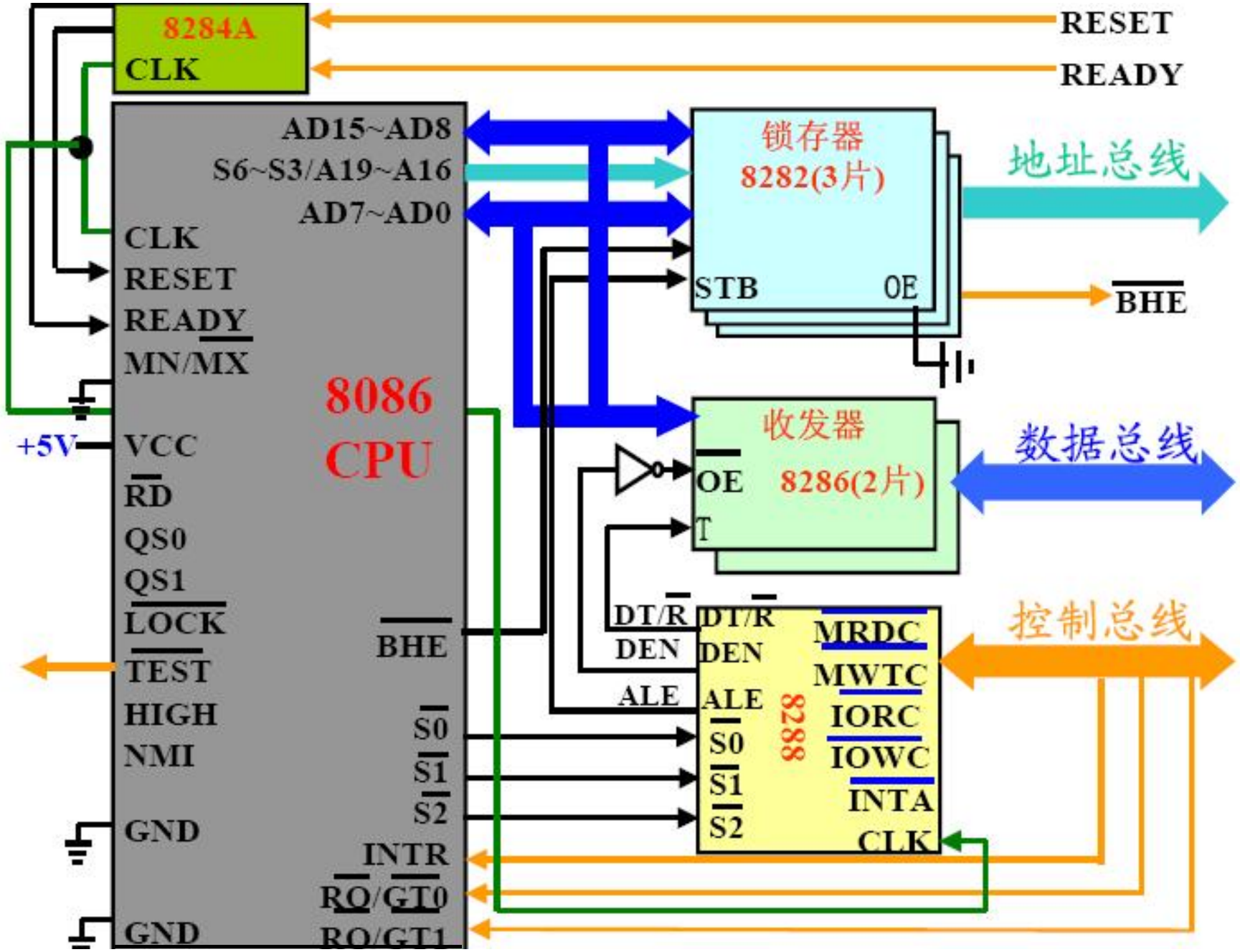

VCC
GND
8284A CLOCK GENERATOR
RES
RDY
WAIT STATE GENERATOR
CLK
READY
RESET
MN/MX
VCC
M/IO
INTA
RD
WR
DT/R
DEN
8086 CPU
ALE
GND
AD0-AD15
A16-A19
BHE
ADDR/DATA
CLK
G
LATCH, 2 OR 3
ADDR
DIR
G
TRANSCEIVER (2)
OPTIONAL FOR INCREASED DATA BUS DRIVE
DATA
BHE
A0
CSOH
CSOL
WE OD
2142 RAM (4)
(2) 1Kx8
(2) 1Kx8
CE
OE
2716-2 PROM (2)
2K x 8
2K x 8
CS
RD WR
MCS-80 PERIPHERAL

8086 最大工作模式

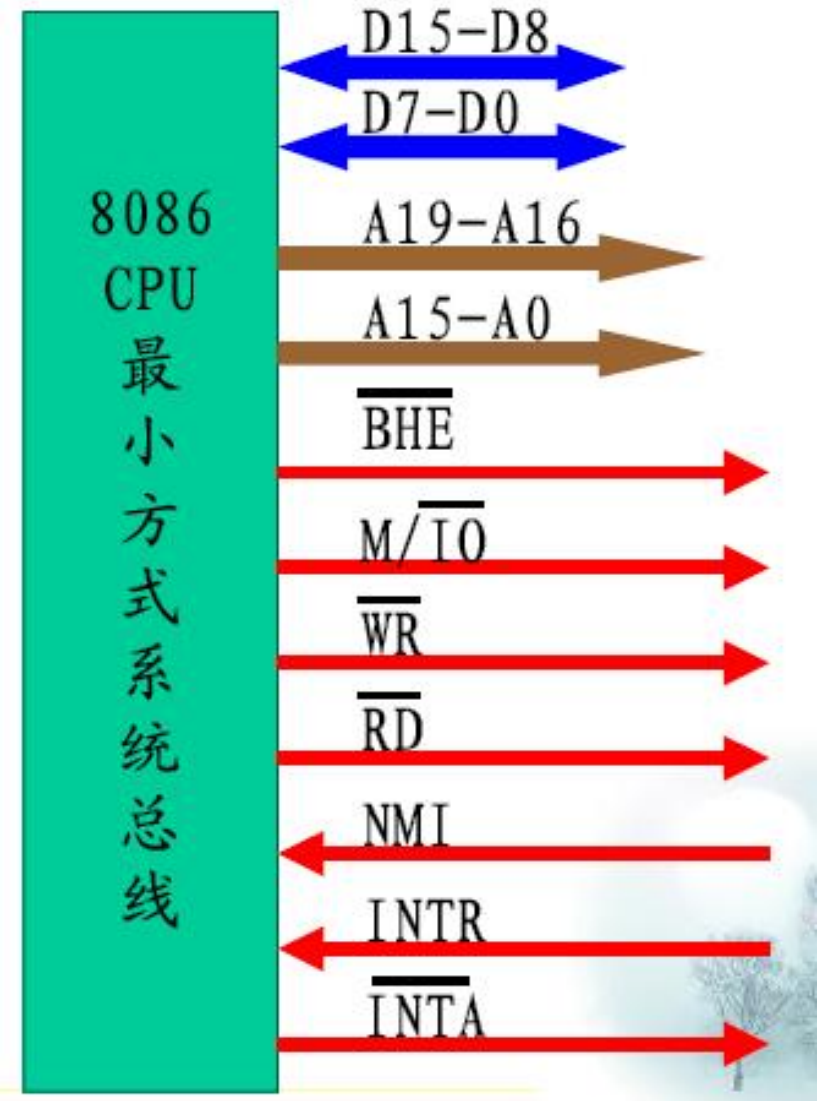

VCC
GND
8284A
CLOCK
GENERATOR
RES
RDY
WAIT
STATE
GENERATOR
MN/MX
GND
CLK
READY
RESET
S0
S1
S2
8086
CPU
LOCK
N.C.
GND
AD0-AD15
A16-A19
ADDR/DATA
BHE
CLK
S0
S1
S2
DEN
DT/R
ALE
8288
BUS
CTRLR
MRDC
MWTC
AMWC
IORC
IOWC
AIOWC
INTA
N.C.
N.C.
CLK
G
LATCH
(2 OR 3)
ADDR
DIR
G
TRANSCEIVER
(2)
DATA
BHE
A0
CSOH
CSOL
WE OD
2142 RAM (4)
(2)
1Kx8
(2)
1Kx8
CE
OE
2716-2 PROM (2)
2K x 8
2K x 8
CS
RD WR
MCS-80
PERIPHERAL

8288对CPU输出的$\overline{S_2}$，$\overline{S_1}$，$\overline{S_0}$三个状态信号译码产生的控制信号如下图所示。

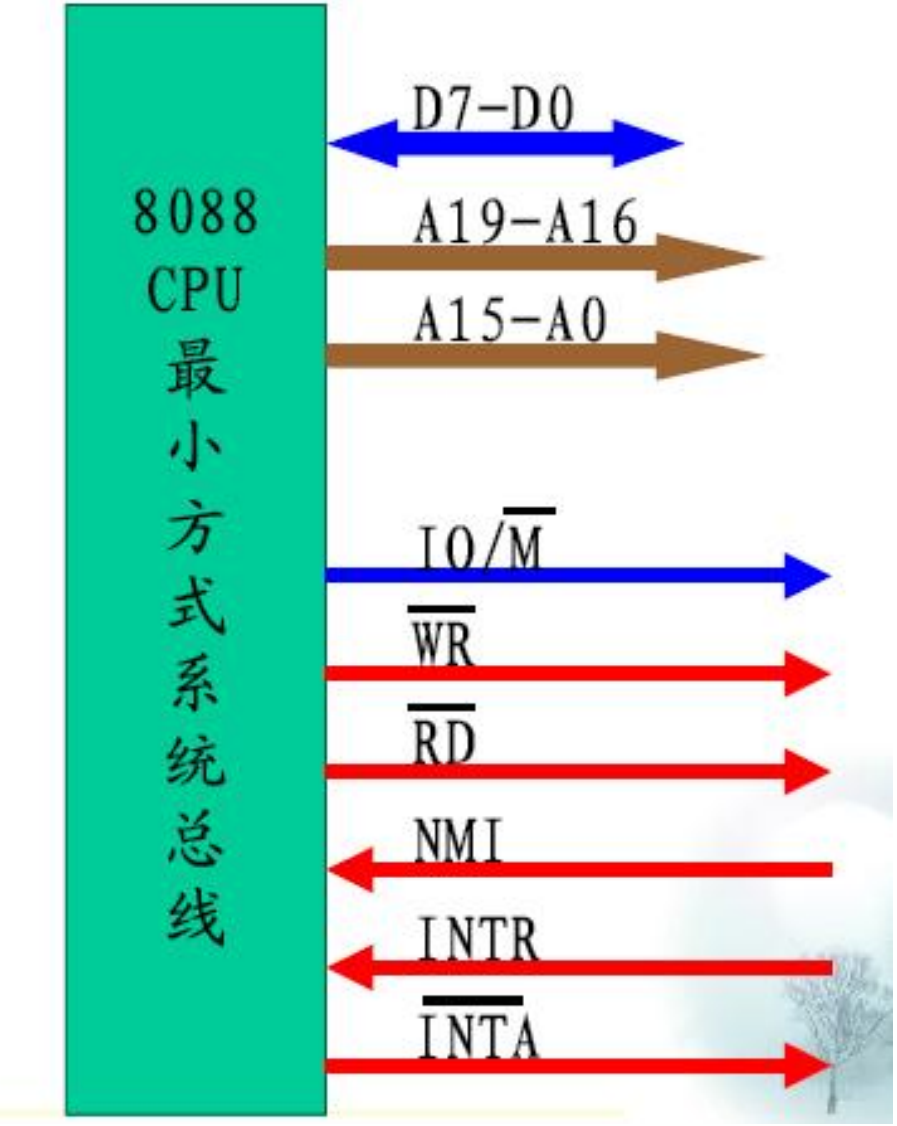



8284A
CLK
RESET
READY
AD15~AD8
S6~S3/A19~A16
AD7~AD0
CLK
RESET
READY
MN/MX
8086
CPU
+5V
VCC
RD
QS0
QS1
LOCK
TEST
HIGH
NMI
GND
GND
BHE
S0
S1
S2
INTR
RQ/GT0
RQ/GT1
锁存器
8282(3片)
STB
OE
地址总线
BHE
收发器
OE
8286(2片)
T
数据总线
DT/R
DEN
ALE
DT/R
DEN
ALE
S0
S1
S2
8288
MRDC
MWTC
IORC
IOWC
INTA
CLK
控制总线

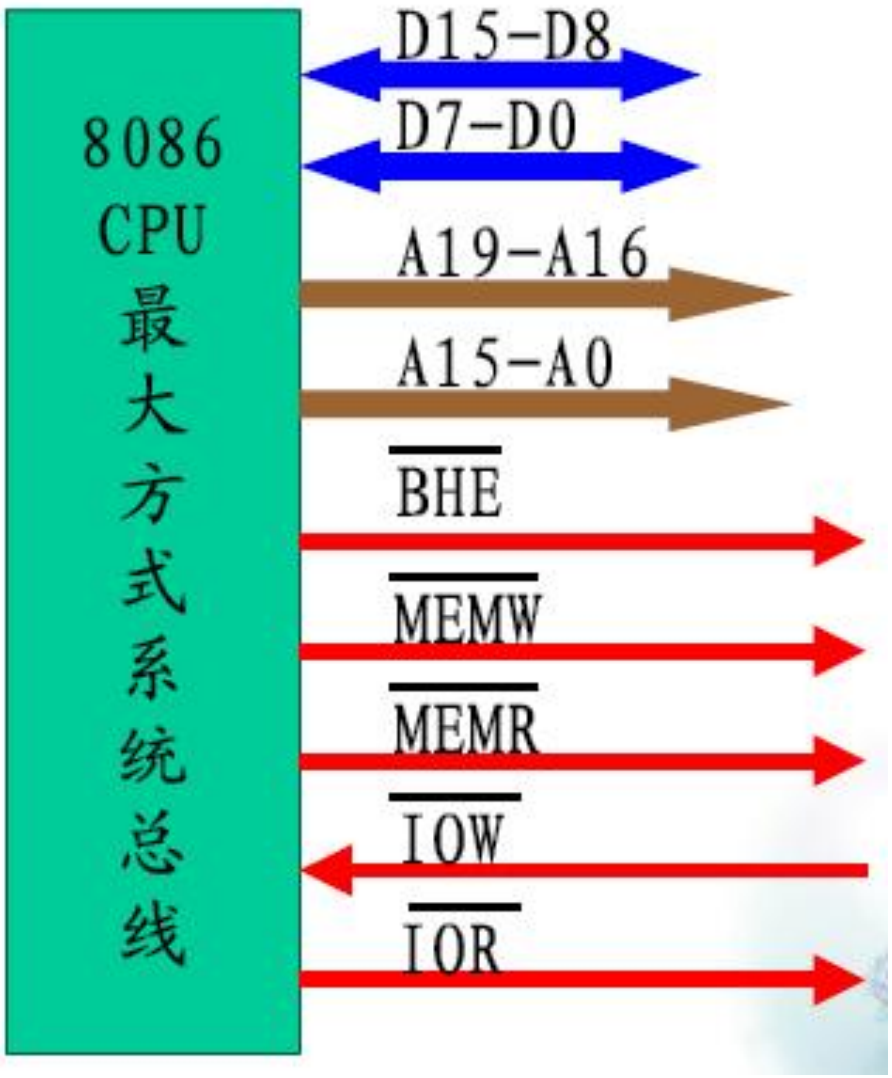
① 8086最小工作方式
系统总线



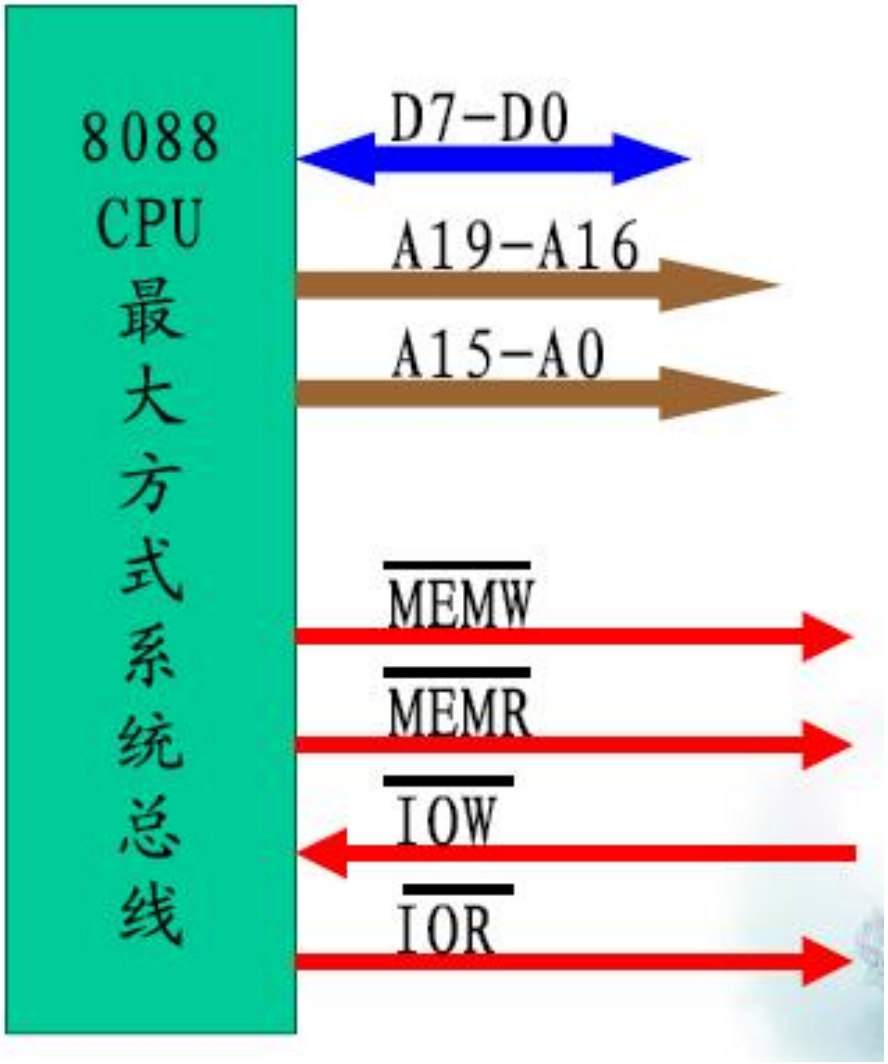
② 8088最小工作方式
系统总线



③ 8086最大工作方式系统总线

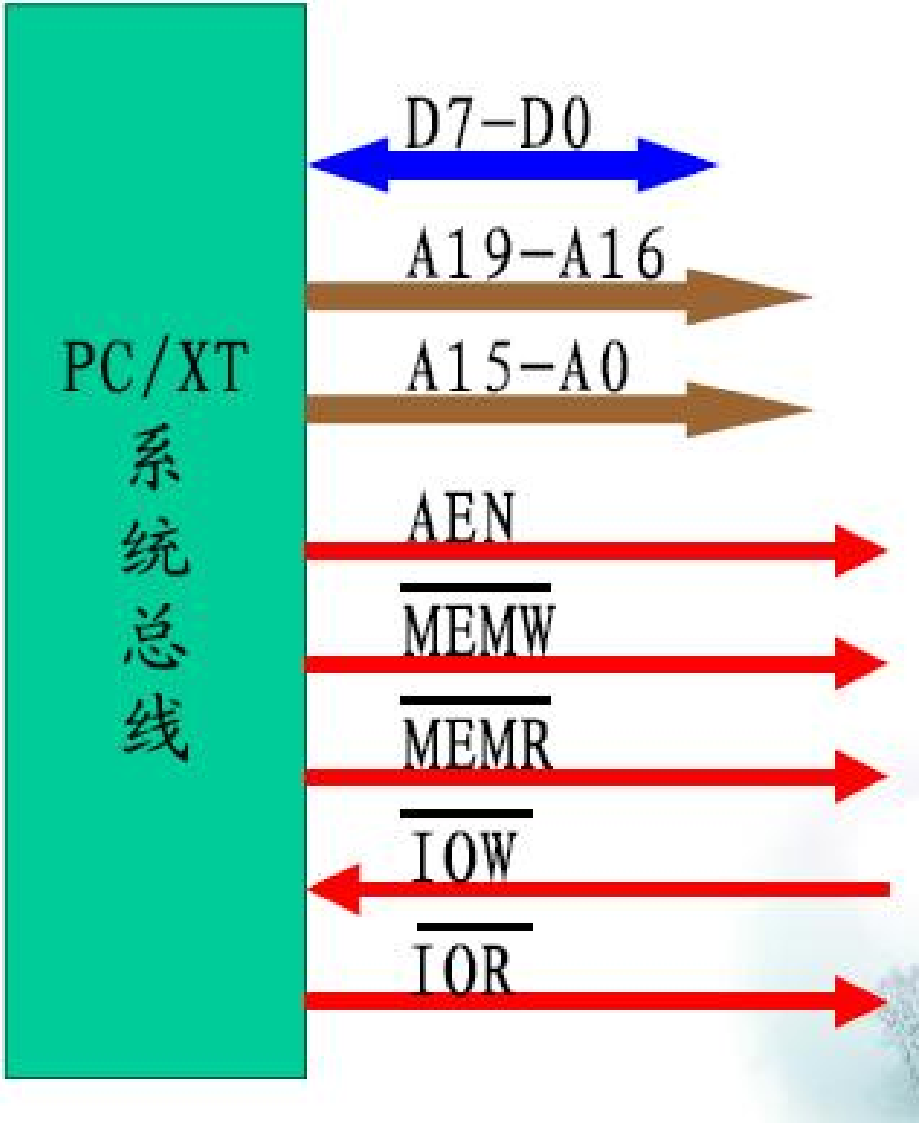


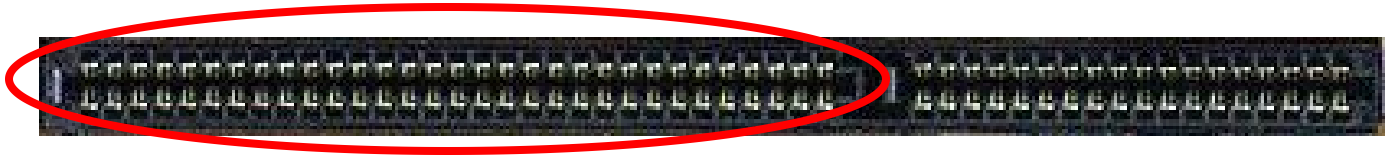
④ 8088最大工作方式
系统总线



⑤ PC/XT系统总线

CPU----8088
最大模式





习　　题

1.在8088 CPU最小工作方式系统中，20位地址全部用74LS373 8位锁存器锁存，数据线上使用74LS245总线收发器，试画出系统总线结构图。